

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12417871
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Lee; Dong Hwan et al.

Coil component

Abstract

A coil component includes a body having an upper surface and a lower surface opposing each other in a thickness direction, a support substrate disposed in the body and having one surface perpendicular to the lower surface of the body, a coil portion disposed on the support substrate, and first and second external electrodes disposed on the lower surface of the body. The body includes an active portion, covering the coil portion, and first and second cover portions, respectively disposed on opposing surfaces of the active portion. The active portion includes a first magnetic metal particle, a second magnetic metal particle having a smaller diameter than the first magnetic metal particle, and a first insulating resin. Each of the first and second cover portions includes a third magnetic metal particle, having a smaller diameter than the first magnetic metal particle, and a second insulating resin.

Inventors: Lee; Dong Hwan (Suwon-si, KR), Cha; Hye Yeon (Suwon-si, KR), Yoon; Chan (Suwon-si, KR), Lee; Dong Jin (Suwon-si, KR), Kim; Hwi Dae (Suwon-si, KR)

Applicant: SAMSUNG ELECTRO-MECHANICS CO., LTD. (Suwon-si, KR)

Family ID: 1000008819375

Assignee: SAMSUNG ELECTRO-MECHANICS CO., LTD. (Suwon-si, KR)

Appl. No.: 17/360154

Filed: June 28, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20220102047 A1	Mar. 31, 2022

Foreign Application Priority Data

KR	10-2020-0125602	Sep. 28, 2020
----	-----------------	---------------

Publication Classification

Int. Cl.: H01F27/02 (20060101); H01F17/04 (20060101); H01F27/25 (20060101); H01F27/255 (20060101); H01F27/29 (20060101)

U.S. Cl.:

CPC H01F27/022 (20130101); H01F17/04 (20130101); H01F27/255 (20130101); H01F27/292 (20130101); H01F2017/048 (20130101)

Field of Classification Search

CPC: H01F (27/022); H01F (17/04); H01F (27/255); H01F (27/292); H01F (2017/048)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
11239019	12/2021	Endo	N/A	H01F 27/2804
12020854	12/2023	Ohkubo	N/A	H01F 27/324
2015/0371752	12/2014	Park	336/200	H01F 27/255
2016/0126004	12/2015	Yang	336/90	H01F 27/292
2016/0163444	12/2015	Choi	336/200	H01F 41/043
2019/0189338	12/2018	Kim	N/A	H01F 27/29
2019/0198221	12/2018	Choi	N/A	H01F 27/36
2020/0090853	12/2019	Yang	N/A	H01F 17/04

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
10-1580399	12/2014	KR	N/A
10-2016-0034802	12/2015	KR	N/A
10-1607026	12/2015	KR	N/A
10-2016-0051541	12/2015	KR	N/A
20160051541	12/2015	KR	N/A
10-2019-0062342	12/2018	KR	N/A
10-2019-0077935	12/2018	KR	N/A
10-2020-0031426	12/2019	KR	N/A

OTHER PUBLICATIONS

Office Action issued in corresponding Korean Patent Application No. 10-2020-0125602 on Oct. 3, 2024, with English translation. cited by applicant

Primary Examiner: Barnes; Malcolm

Attorney, Agent or Firm: MORGAN, LEWIS & BOCKIUS LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

(1) The present application claims the benefit of priority to Korean Patent Application No. 10-2020-0125602, filed on Sep. 28, 2020 in the Korean Intellectual Property Office, the entire disclosure of which is incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to a coil component.

BACKGROUND

(3) An inductor, a coil component, is a typical passive electronic component used in electronic devices, along with a resistor and a capacitor.

(4) As electronic devices gradually gain higher performance and become smaller, the number of electronic components used in electronic devices is increased, while being miniaturized.

(5) In the case of a thin film type coil component, a body is formed by laminating a magnetic composite sheet, in which magnetic metal particles are dispersed in an insulating resin, on a substrate on which a coil portion is formed through a plating process, and curing the laminated magnetic composite sheet, and an external electrode is formed on a surface of the body.

SUMMARY

(6) An aspect of the present disclosure is to provide a coil component, in which a coil portion is disposed to be perpendicular to a lower surface of a body, capable of readily preventing short-circuits between external electrodes.

(7) Another aspect of the present disclosure is to provide a coil component, in which a coil portion is disposed to be perpendicular to a lower surface of a body, capable of increasing an effective volume of a magnetic material.

(8) According to an aspect of the present disclosure, a coil component includes a body having an upper surface and a lower surface opposing each other in a thickness direction of the body, a support substrate disposed in the body and having one surface perpendicular to the lower surface of the body, a coil portion disposed on at least one surface of the support substrate, and a first external electrode and a second external electrode disposed on the lower surface of the body to be spaced apart from each other and respectively connected to the coil portion. The body includes an active portion, covering the coil portion, and first and second cover portions, respectively disposed on an upper surface and a lower surface of the active portion opposing each other in the thickness direction. The active portion includes a first magnetic metal particle, a second magnetic metal particle having a smaller diameter than the first magnetic metal particle, and a first insulating resin. Each of the first and second cover portions includes a third magnetic metal particle, having a smaller diameter than the first magnetic metal particle, and a second insulating resin.

(9) According to an aspect of the present disclosure, a coil component includes a body having a lower surface; a support substrate disposed in the body and having one surface perpendicular to the lower surface of the body; and a coil portion disposed on at least one surface of the support substrate. The body includes an active portion, covering the coil portion, and first to fourth cover portions, respectively disposed on an upper surface of the body, the lower surface of the body, one side surface of the body, and the other side surface of the body. Each of the active portion and the first to fourth cover portions includes magnetic metal particles and an insulating resin. A filling rate of the magnetic metal particles of the active portion is greater than a filling rate of the magnetic metal particles of each of the first to fourth cover portions.

(10) According to an aspect of the present disclosure, a coil component includes a body having an upper surface and a lower surface opposing each other; a coil portion disposed in the body vertically with respect to the lower surface of the body; and a first external electrode and a second external electrode respectively including pad portions disposed on the lower surface of the body to be spaced apart from each other and connected to the coil portion. An average size of metal particles dispersed in a portion, in contact with one of the pad portions, of the body, is less than an

average size of metal particles dispersed in another portion, spaced apart from outer surfaces of the body, of the body.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) The above and other aspects, features, and advantages of the present disclosure will be more clearly understood from the following detailed description, taken in conjunction with the accompanying drawings.
- (2) FIG. 1 is a schematic perspective view of a coil component according to an exemplary embodiment of the present disclosure.
- (3) FIG. 2 is a cross-sectional view taken along line I-I' of FIG. 1.
- (4) FIG. 3 is a cross-sectional view taken along line II-II' of FIG. 1.
- (5) FIG. 4 is a schematic perspective view of another coil component according to an exemplary embodiment of the present disclosure.
- (6) FIG. 5 is a cross-sectional view taken along line III-III' of FIG. 4.
- (7) FIG. 6 is a cross-sectional view taken along line IV-IV' of FIG. 4.
- (8) FIG. 7 is a schematic perspective view of another coil component according to an exemplary embodiment of the present disclosure.
- (9) FIG. 8 is a cross-sectional view taken along line V-V' of FIG. 7.
- (10) FIG. 9 is a cross-sectional view taken along line VI-VI' of FIG. 7.

DETAILED DESCRIPTION

- (11) The terms used in the description of the present disclosure are used to describe a specific embodiment, and are not intended to limit the present disclosure. A singular term includes a plural form unless otherwise indicated. The terms “include,” “comprise,” “is configured to,” etc. of the description of the present disclosure are used to indicate the presence of features, numbers, steps, operations, elements, parts, or combination thereof, and do not exclude the possibilities of combination or addition of one or more additional features, numbers, steps, operations, elements, parts, or combination thereof. Also, the terms “disposed on,” “positioned on,” and the like, may indicate that an element is positioned on or beneath an object, and does not necessarily mean that the element is positioned above the object with reference to a direction of gravity.
- (12) Terms such as “coupled to,” “combined to,” and the like, may not only indicate that elements are directly and physically in contact with each other, but also include the configuration in which another element is interposed between the elements such that the elements are also in contact with the other component.
- (13) Sizes and thicknesses of elements illustrated in the drawings are indicated as examples for ease of description, and the present disclosure are not limited thereto.
- (14) In the drawings, an L direction is a first direction or a length (longitudinal) direction, a W direction is a second direction or a width direction, a T direction is a third direction or a thickness direction.
- (15) Hereinafter, a coil component according to an exemplary embodiment of the present disclosure will be described in detail with reference to the accompanying drawings. Referring to the accompanying drawings, the same or corresponding components may be denoted by the same reference numerals, and overlapped descriptions will be omitted.
- (16) In electronic devices, various types of electronic components may be used, and various types of coil components may be used between the electronic components to remove noise, or for other purposes.
- (17) In other words, in electronic devices, a coil component may be used as a power inductor, a high frequency (HF) inductor, a general bead, a high frequency (GHz) bead, a common mode filter,

and the like.

(18) FIG. 1 is a schematic perspective view of a coil component according to an exemplary embodiment. FIG. 2 is a cross-sectional view taken along line I-I' of FIG. 1, and FIG. 3 is a cross-sectional view taken along line II-II' of FIG. 1. For ease of description, in FIG. 2, an internal structure of the coil component is illustrated based on the cross-section taken along line I-I' of FIG. 1.

(19) Referring to FIGS. 1 to 3, a coil component **1000** according to an exemplary embodiment may include a body **100**, a support substrate **200**, a coil portion **300**, and external electrodes **410** and **420**. The body **100** may include an active portion C and first to fourth cover portions **121**, **122**, **123** and **124**.

(20) The body **100** may form an exterior of the coil component **1000** according to the present embodiment, and may have the support substrate **200** and the coil portion **300** embedded therein.

(21) The body **100** may be formed to have an overall hexahedral shape.

(22) Based on FIGS. 1 to 3, the body **100** has a first surface **101** and a second surface **102** opposing each other in a length direction L, a third surface **103** and a fourth surface **104** opposing each other in a width direction W, and a fifth surface **105** and a sixth surface **106** opposing each other in a thickness direction T. Each of the first to fourth surfaces **101**, **102**, **103**, and **104** of the body **100** may correspond to a wall surface of the body **100** connecting the fifth surface **105** and the sixth surface **106** of the body **100**. Hereinafter, both end surfaces of the body **100** may refer to the first surface **101** and the second surface **102** of the body **100**, respectively, and both side surfaces of the body **100** may refer to the third surface **103** and the fourth surface **104** of the body **100**, respectively. In addition, a lower surface and an upper surface of the body **100** may refer to the sixth surface **106** and the fifth surface **105**, respectively.

(23) The body **100** may be formed such that the coil component **1000**, including the external electrodes **410** and **420** to be described later, has a length of 1.0 mm, a width of 0.5 mm, and a thickness of 0.6 mm or a length of 1.6 mm, a width of 0.8 mm, and a thickness of 1.0 mm, but is not limited thereto. Since the above-described sizes of the coil component **1000** are merely illustrative, cases in which a size of the coil component **1000** are smaller or larger than the above-mentioned dimensions may be not excluded from the scope of the present disclosure.

(24) The above-described length of the coil component **1000** may refer to a maximum value, among lengths of a plurality of segments, connecting outermost boundary lines of the body **100**, among outermost boundary lines of the coil component **1000** illustrated in a cross-sectional image, and parallel to a length (L) direction of the body **100**, based on an optical microscope or scanning electron microscope (SEM) image for a cross-section of the body **100** in a length-thickness (L-T) direction in a central portion of the body **100** in a width (W) direction. Alternatively, the length of the coil component may refer to arithmetic means of lengths of at least three segments, among a plurality of segments connecting outermost boundary lines of the coil component **1000** illustrated in the cross-sectional image, and parallel to the length (L) direction of the body **100**.

(25) The above-described thickness of the coil component **1000** may refer to a maximum value, among thicknesses of a plurality of segments, connecting outermost boundary lines of the body **100**, among outermost boundary lines of the coil component **1000** illustrated in a cross-sectional image, and parallel to a thickness (T) direction of the body **100**, based on an optical microscope or scanning electron microscope (SEM) image for a cross-section of the body **100** in a length-thickness (L-T) direction in a central portion of the body **100** in a width (W) direction.

Alternatively, the thickness of the coil component may refer to arithmetic means of thicknesses of at least three segments, among a plurality of segments connecting outermost boundary lines of the coil component **1000** illustrated in the cross-sectional image, and parallel to the thickness (T) direction of the body **100**.

(26) The above-described width of the coil component **1000** may refer to a maximum value, among widths of a plurality of segments, connecting outermost boundary lines of the body **100**, among

outermost boundary lines of the coil component **1000** illustrated in a cross-sectional image, and parallel to a width (W) direction of the body **100**, based on an optical microscope or scanning electron microscope (SEM) image for a cross-section of the body **100** in a width-thickness (W-T) direction in a central portion of the body **100** in a length (L) direction. Alternatively, the width of the coil component may refer to arithmetic means of widths of at least three segments, among a plurality of segments connecting outermost boundary lines of the coil component **1000** illustrated in the cross-sectional image, and parallel to the width (W) direction of the body **100**.

(27) Alternatively, each of the length, the width, and the thickness of the coil component **1000** may be measured by a micrometer measurement method. In the micrometer measurement method, measurement may be performed by setting a zero point using a micrometer (instrument) with gauge repeatability and reproducibility (R&R), inserting the coil component **1000** between tips of the micrometer, and turning a measurement lever of the micrometer. When the length of the coil component **1000** is measured by a micrometer measurement method, the length of the coil component **1000** may refer to a value measured once or an arithmetic mean of values measured two or more times. This may be equivalently applied to the width and the thickness of the coil component **1000**.

(28) The body **100** may include magnetic metal particles (or metal powder particles) **P1**, **P2**, **P3**, and **P4** and insulating resins **R1**, **R2**, and **R3**. The body **100** is formed by laminating one or more magnetic composite sheets including insulating resins **R1**, **R2**, and **R3** and magnetic metal particles **P1**, **P2**, **P3**, and **P4** dispersed in the insulating resins **R1**, **R2**, and **R3**.

(29) The body **100** includes an active portion **110** and cover portions **121**, **122**, **123**, and **124**.

(30) Specifically, the active portion **110** may cover the coil portion **300** to be described later, and may include a first magnetic metal particle **P1**, a second magnetic metal particle **P2** having a smaller diameter than the first magnetic metal particle **P1**, and a first insulating resin **R1**. The first cover portion **121** may be disposed on an upper surface of the active portion **110** and may include a third magnetic metal particle **P3**, having a smaller diameter than the first magnetic metal particle **P1**, and a second insulating resin **R2**. The second cover portion **122** may be disposed on a lower surface of the active portion **110** and may include a third magnetic metal particle **P3**, having a smaller diameter than the first magnetic metal particle **P1**, and a second insulating resin **R2**. The third and fourth cover portions **123** and **124** may be disposed on side surfaces of the active portion **110** facing each other in a width direction, respectively. Each of the third and fourth cover portions **123** and **124** may include a fourth magnetic metal particle **P4**, having a smaller diameter particle than the first magnetic metal particle **P1**, and a third insulating resin **R3**. In the present specification, a diameter (or a size) or an average diameter (or an average size) of a magnetic metal particle may refer to a particle size distribution represented by **D50** or **D90**, measured, for example, based on an optical microscope or scanning electron microscope (SEM) image of a cross-section of the body **100**. Other methods and/or tools appreciated by one of ordinary skill in the art, even if not described in the present disclosure, may also be used.

(31) The active portion **110** may be formed by laminating a first magnetic composite sheet, including the first and second magnetic metal powder particles **P1** and **P2** and the first insulating resin **R1**, on one surface and the other surface of the support substrate **200** to be described later in the width direction W. Since the active portion **110** includes the first magnetic metal powder particle **P1** having a relatively large diameter and the second magnetic metal powder particle **P2** having a relatively small particle size, the active portion **110** may have a relatively high filling rate of the magnetic metal powder particles **P1** and **P2**, as compared with the first to fourth cover portions **121**, **122**, **123**, and **124** including only the third and fourth magnetic metal powder particles **P3** and **P4** having a relatively small grain size, as compared with the first magnetic metal powder **P1**. The term “filling rate of a magnetic metal powder” may refer to an overall volume or an overall mass occupied by a magnetic metal powder particle within a unit volume (for example, a unit volume of a length of 10 μm * a width of 10 μm * a thickness of 10 μm) of each of the active

portion **110** and the cover portions **121**, **122**, **123**, and **124**. Hereinafter, for ease of description, a description will be provided under the assumption that the magnetic metal powders **P1** and **P2** of the active portion **110** include only the first magnetic metal powder particle **P1** and the second magnetic metal powder particle **P2** having different particle sizes, but the present disclosure is not limited thereto. For example, as another non-limiting example of the present disclosure, the magnetic metal powder particles **P1**, **P2**, and **P3** of the active portion **110** may include three types of powder particles having different diameters.

(32) The first and second cover portions **121** and **122** may be disposed on upper and lower surfaces of the active portion **110** facing each other in the thickness direction **T**, respectively. In general, a plurality of individual components may be formed in a batch by manufacturing a coil bar, in which a plurality of bodies are connected to each other, and then separating the plurality of bodies from each other through a dicing process. In the case of a horizontally disposed component in which one surface of a support substrate is disposed to be parallel to a mounting surface of the component, a dicing surface does not constitute the upper and lower surfaces of the body in an individual component. On the other hand, in the case of a vertically disposed component in which one surface of a support substrate is disposed to be perpendicular to mounting surface, upper and lower surfaces of the body correspond to cut surfaces formed by dicing. For this reason, at least a portion of a magnetic metal powder particle included in the body may be cut by a dicing blade to be exposed to the upper and lower surfaces of the body. When an external electrode is formed by plating in such a state, a magnetic metal powder particle cut to expose a conductive core may cause plating spreading to occur on the upper surface and the lower surface of the body, resulting in short-circuits of the external electrode. In the present embodiment, the above-described issue may be addressed by providing the first and second cover portions **121** and **122** on the upper and lower surfaces of the active portion **110**, i.e., dicing surfaces. For example, in the case of the present embodiment, at least a portion of the first and second magnetic metal powder particles **P1** and **P2** may be cut, and thus, cut surfaces may be exposed to the upper and lower surfaces of the active portion **110** (in particular, the magnetic metal powder particle **P1** having a relatively large grain size is often cut to be exposed). A first cover portion **121** and a second cover portions **122**, including a third magnetic metal powder particle **P3** and a second insulating resin **R2**, may be disposed on the upper and lower surfaces of the active portion **110**, respectively, to cover the conductive core of the magnetic metal powder particle exposed to the upper and lower surfaces of the active portion **110**. For example, the first and second cover portions **121** and **122** may cover the upper and lower surfaces of the active portion **110**, i.e., cut surfaces of the active portion **110**. For this reason, the conductive cores of the magnetic metal powder particles **P1**, **P2**, and **P3** may not be exposed to the upper and lower surfaces of the body **100**. Accordingly, plating spreading, occurring on the upper and lower surfaces of the body **100** when plating the external electrodes **410** and **420**, may be reduced. Since the first and second cover portions **121** and **122** include a third magnetic metal powder particle **P3** having a diameter smaller than the first magnetic metal powder particle **P1**, surface roughness of the surface of the first and second cover portions **121** and **122** may be reduced. Pad portions **411** and **421** and band portions **413** and **423** of the external electrodes **410** and **420** to be described later may be disposed on the upper and lower surfaces of the body **100**, and an exterior defect of the external electrodes **410** and **420** may be reduced due to relatively low surface roughness of the upper and lower surfaces of the body **100**. In addition, the pad portions **411** and **421** and the band portions **413** and **423** of the external electrode may be formed to be uniform. The diameters of the third magnetic metal powder particle **P3** and the diameter of the second magnetic metal powder particle **P2** may be the same. In this case, ease of process may be increased. The second insulating resin **R2** may be different from the first insulating resin **R1**. However, the present disclosure is not limited thereto and the first and second insulating resins **R1** and **R2** may include, for example, the same resin such as an epoxy resin, whereby bonding force between the active portion **110** and the first and second cover portions **121** and **122** may be

improved. Since curing of the active portion **110** has been finished when the first and second cover portions **121** and **122** were formed, an interface may be formed between each of the first and second cover portions **121** and **122** and the active portion **110**. The first and second cover portions **121** and **122** may be formed by a sheet lamination method or a paste applying method, but the present disclosure is not limited thereto.

(33) The third and fourth cover portions **123** and **124** may be disposed on both surfaces of the active portion **110** facing in the width direction T, respectively. For example, based on the direction of FIG. 1, the third cover portion **123** may be disposed on a front surface of the active portion **110** to form the third surface **103** of the body **100**, and the fourth cover portion **124** may be disposed on a rear surface of the active portion **110** to form the fourth surface **104** of the body **100**. The third and fourth cover portions **123** and **124** may be formed in a process of laminating the first magnetic composite sheet for forming the active portion **110** by providing a second magnetic composite sheet, including a fourth magnetic metal powder particle P4 having a smaller diameter than the first magnetic metal powder particle P1 and a third insulating layer R3, on the first magnetic composite sheet disposed on an outermost side of the width direction W and then laminating the first and second magnetic composite sheets. For example, unlike the first and second cover portions **121** and **122**, the third and fourth cover portions **123** and **124** may be formed together in the same process as the active portion **110**. Since the first and second cover portions **121** and **122** form the third and fourth surfaces **103** and **104** of the body **100**, an exterior defect of the body **100** may be reduced. A diameter of the fourth magnetic metal powder particle P4 may be the same as a diameter of the third magnetic metal powder particle P3 or the diameter of the second magnetic metal powder particle P2. The third insulating resin R3 may include the same resin as the first insulating resin R1, but the present disclosure is not limited thereto. For the above-described reasons of process, the upper surface of the active portion **110** and the upper surface of each of the third and fourth cover portions **123** and **124** may be coplanar with each other, and the lower surface of the active portion **110** and the lower surface of each of the third and fourth cover portions **123** and **124** may be coplanar with each other. In addition, for the above-described reasons of process, at least a portion of each of the upper and lower surfaces of the third and fourth cover portions **123** and **124**, coplanar with each of the upper and lower surfaces of the active portion **110**, may be covered.

(34) Each of the first to fourth cover portions **121**, **122**, **123**, and **124** may have a thickness of 20 μm to 30 μm . The thickness of the first and second cover portions **121** and **122** may refer to a length (dimension) of the first and second cover portions **121** and **122** in the thickness direction T, and the thickness of the third and fourth cover portions **123** and **124** may refer to a length (dimension) of the third and fourth cover portions **123** and **124** in the width direction W. For example, when the thickness of the first cover portion **121** is less than 20 μm , it may be difficult to secure insulation between the external electrodes **410** and **420**, and there is a probability that short-circuits may occur between the external electrodes **410** and **420**. For example, when the thickness of the first cover portion **121** is greater than 30 μm , the thickness of the entire component may be increased.

(35) Each of the magnetic metal particles P1, P2, P3, and P4 may include one or more selected from the group consisting of iron (Fe), silicon (Si), chromium (Cr), cobalt (Co), molybdenum (Mo), aluminum (Al), niobium (Nb), copper (Cu), and nickel (Ni). For example, the magnetic metal particle may be at least one or more of a pure iron particle, a Fe—Si-based alloy particle, a Fe—Si—Al-based alloy particle, a Fe—Ni-based alloy particle, a Fe—Ni—Mo-based alloy particle, a Fe—Ni—Mo—Cu-based alloy particle, a Fe—Co-based alloy particle, a Fe—Ni—Co-based alloy particle, a Fe—Cr-based alloy particle, a Fe—Cr—Si-based alloy particle, a Fe—Si—Cu—Nb-based alloy particle, a Fe—Ni—Cr-based alloy particle, and a Fe—Cr—Al-based alloy particle.

(36) The magnetic metal particles P1, P2, P3, and P4 may be amorphous or crystalline. For example, the magnetic metal particles P1, P2, P3, and P4 may be a Fe—Si—B—Cr-based

amorphous alloy particle, but is not limited thereto. Each of the magnetic metal particles P1, P2, P3, and P4 may have an average diameter of about 0.1 μm to 30 μm , but is not limited thereto.

(37) In each of the magnetic metal particles P1, P2, P3, and P4, a conductive core, the above-described material, and an insulating coating layer, coating a surface of the core, may be disposed.

(38) Each of the insulating resins R1, R2, and R3 may include epoxy, polyimide, liquid crystal polymer, or the like, in a single or combined form, but is not limited thereto.

(39) The body 100 may have a core C penetrating through the support substrate 200 and the coil portion 300 to be described later. The core C may be formed by filling a through-hole of the coil portion 300 with a magnetic composite sheet, but the present disclosure is not limited thereto.

(40) The support substrate 200 may be disposed within the body 100, and may have one surface perpendicular to the lower surface 106 of the body 100. The support substrate 200 may be configured to support the coil portion 300 to be described later. The sentence “one surface of the support substrate 200 is perpendicular to the lower surface 106 of the body 100” may mean that, as an example, one surface of the support substrate 200 and the lower surface 106 of the body 100 form an angle of 80 to 100 degrees, while including meaning of perpendicularity in mathematical sense. Element A such as an object, a plane, or a line/direction being perpendicular to or vertical to element B such as another object, another plane, or another line/direction may mean that, as an example, element A and element B form an angle of 80 to 100 degrees, while including meaning of perpendicularity in mathematical sense. In one example, a winding axis of the coil portion 300 may be parallel to the lower surface 106 of the body 100. The sentence “a winding axis of the coil portion 300 may be parallel to the lower surface 106 of the body 100” may mean that, as an example, a winding axis and the lower surface 106 of the body 100 form an angle of -10 to 10 degrees, while including meaning of parallel in mathematical sense. Element C such as an object, a plane, or a line/direction being parallel to element D such as another object, another plane, or another line/direction may mean that, as an example, element C and element D form an angle of -10 to 10 degrees, while including meaning of parallel in mathematical sense.

(41) The support substrate 200 may include an insulating material, for example, a thermosetting insulating resin such as an epoxy resin, a thermoplastic insulating resin such as polyimide, or a photosensitive insulating resin, or the support substrate 200 may include an insulating material in which a reinforcing material such as a glass fiber or an inorganic filler is impregnated with an insulating resin. For example, the support substrate 200 may include an insulating material such as prepreg, Ajinomoto Build-up Film (ABF), FR-4, a bismaleimide triazine (BT) film, a photoimageable dielectric (PID) film, a copper clad laminate (CCL), or the like, but the present disclosure is not limited thereto.

(42) The inorganic filler may be at least one or more selected from the group consisting of silica (SiO.sub.2), alumina (Al.sub.2O.sub.3), silicon carbide (SiC), barium sulfate (BaSO.sub.4), talc, mud, a mica particle, aluminum hydroxide (Al(OH).sub.3), magnesium hydroxide (Mg(OH).sub.2), calcium carbonate (CaCO.sub.3), magnesium carbonate (MgCO.sub.3), magnesium oxide (MgO), boron nitride (BN), aluminum borate (AlBO.sub.3), barium titanate (BaTiO.sub.3), and calcium zirconate (CaZrO.sub.3).

(43) When the support substrate 200 is formed of an insulating material including a reinforcing material, the support substrate 200 may provide more improved rigidity. When the support substrate 200 is formed of an insulating material including no glass fiber, the support substrate 200 is advantageous for thinning the coil component 1000. In addition, the effective volume of the coil portion 300 and/or the magnetic material may be increased, based on a component having the same volume, to improve component characteristics. When the support substrate 200 is formed of an insulating material including a photosensitive insulating resin, the number of processes for forming the coil portion 300 may be decreased. Therefore, it may be advantageous in reducing production costs, and a fine via may be formed.

(44) The coil portion 300 may be disposed on at least one surface of the support substrate 200. The

coil portion **300** may be embedded in the body **100** to express characteristics of the coil component **1000**. For example, when the coil component **1000** is used as a power inductor, the coil portion **300** may store an electric field as a magnetic field to maintain an output voltage, serving to stabilize power of an electronic device.

(45) The coil portion **300** may be formed on at least one of both surfaces of the support substrate **200** facing each other, and may form at least one turn. The coil portion **300** may be disposed on one surface and the other surface of the support substrate **200** of the body **100** facing each other in the width direction W.

(46) The coil portion **300** may include coil patterns **311** and **312**, a via **321**, and lead-out patterns **331** and **332**. Specifically, each of the first coil pattern **311** and the second coil pattern **312** may be in the form of a planar spiral in which at least one turn is formed around the core **110**. For example, based on the direction of FIG. 1, the first coil pattern **311** may form at least one turn around the core **110** on the rear surface of the support substrate **200**. The second coil pattern **312** may form at least one turn around the core C on the front surface of the support substrate **200**. The first lead-out pattern **331** may be disposed on the rear surface of the support substrate **200** to be connected to the first coil pattern **311** and to be exposed to the first surface **101** of the body **100**. The second lead-out pattern **332** may be disposed on the front surface of the support substrate **200** to be connected to the second coil pattern **312** and to be exposed to the second surface **102** of the body **100**. The via **321** may penetrate through the support substrate **200** to connect innermost end portions of the first and second coil patterns **311** and **312** to each other. Thus, the coil portion **300** may function as a single coil connected overall.

(47) At least one of the coil patterns **311** and **312**, the vias **321**, and the lead-out patterns **331** and **332** may include at least one conductive layer.

(48) As an example, when the second coil pattern **312**, the via **321**, and the second lead-out pattern **332** are formed on the front surface of the support substrate **200** by plating, each of the second coil pattern **312**, the via **321**, and the second lead-out pattern **332** may include a seed layer and an electroplating layer. The seed layer may be formed by electroless plating or vapor deposition such as sputtering. Each of the seed layer and the electroplating layer may have a single-layer structure or a multilayer structure. An electroplating layer having a multilayer structure may be formed to have a conformal film structure in which one electroplating layer is covered with another electroplating layer, or another electroplating layer is laminated on only one surface of one electroplating layer. It may be formed in a shape. The seed layer of the second coil pattern **312**, the seed layer of the via **321**, and the seed layer of the second lead-out pattern **332** may be formed to be integrated with each other, such that a boundary may not be formed therebetween, but the present disclosure is not limited thereto. The electroplating layer of the second coil pattern **312**, the electroplating layer of the via **321**, and the electroplating layer of the second lead-out pattern **332** may be formed to be integrated with each other, such that a boundary may not be formed therebetween, but the present disclosure is not limited thereto.

(49) Each of the coil patterns **311** and **312**, the via **321**, and the lead-out patterns **331** and **332** may include a conductive material such as copper (Cu), aluminum (Al), silver (Ag), tin (Sn), gold (Au), nickel A conductive material such as (Ni), lead (Pb), titanium (Ti), chromium (Cr), molybdenum (Mo), or alloys thereof, but the present disclosure is not limited thereto.

(50) In the present embodiment, since the coil portion **300** is disposed to be perpendicular to the sixth surface **106** of the body **100**, a mounting surface, a mounting area may be reduced while maintaining the volume of the body **100**. Therefore, a larger number of electronic components may be mounted on a mounting board having the same area. In addition, since the coil portion **300** is disposed to be perpendicular to the sixth surface **106** of the body **100**, the mounting surface, a direction of magnetic flux induced by the coil portion **300** may be disposed to be parallel to the sixth surface **106** of the body **100**. For this reason, noise induced to the mounting surface of the mounting board may be relatively reduced.

(51) An insulating layer IF may be provided to insulate the coil portion **300** from the body **100**, and may include a known insulating material such as parylene, but the present disclosure is not limited thereto. The insulating layer IF may be formed by vapor deposition. However, the present disclosure is not limited thereto, and the insulating layer IF may be formed by laminating an insulating film on both surfaces of the support substrate **200**.

(52) The external electrodes **410** and **420** may electrically connect the coil component **1000** according to the present embodiment to a printed circuit board (PCB), or the like, when the coil component **1000** is mounted on the PCB, or the like. As an example, the coil component **1000** may be mounted such that the sixth surface **106** of the body **100** faces an upper surface of the PCB, and the external electrodes **410** and **420**, disposed on the sixth surface **106** of the body **100** to be spaced apart from each other, and a connection portion of the PCB may be electrically connected to each other.

(53) The external electrodes **410** and **420** may be connected to the lead-out patterns **331** and **332**, disposed on the sixth surface **106** of the body **100** to be spaced apart from each other, respectively. Specifically, the first external electrode **410** may include a first pad portion **411** disposed on the sixth surface **106** of the body **100**, a first extension portion **412** extending from the first pad portion **411** to the first surface **101** of the body **100**, and a first band portion **413** extending from the first extension portion **412** to the fifth surface **105** of the body **100**. The first extension portion **412** may be in contact with and connected to the first lead-out pattern **331** exposed to the first surface **101** of the body **100**. The second external electrode **420** may include a second pad portion **421** disposed on the sixth surface **106** of the body **100**, a second extension portion **422** extending from the second pad portion **421** onto the second surface **102** of the body **100**, and a second band portion **423** extending from the second extension portion **422** onto the fifth surface **105** of the body **100**. The second extension portion **422** may be in contact with and connected to the second lead-out pattern **332** exposed to the second surface **102** of the body **100**. For example, each of the first and second external electrodes **410** and **411** may be formed to have an overall $\sqcap$ -shape. For example, each of the pad portions **411** and **421** and the band portions **413** and **423** may have a thickness of 20 μm , and each of the extension portions **412** and **422** may have a thickness of 30 μm , but the present disclosure is limited thereto. The phrase “thickness of each of the pad portions **411** and **421** and the band portions **413** and **423**” may refer to a length (dimension) of each of the pad portions **411** and **421** and the band portions **413** and **423** in the thickness direction T. The phrase “thickness of each of the extension portions **412** and **422**” may refer to a length (dimension) of each of the extension portions **412** and **422** in the width direction W.

(54) The external electrodes **410** and **420** may be formed of a conductive material such as copper (Cu), aluminum (Al), silver (Ag), tin (Sn), gold (Au), nickel (Ni), lead (Pb), chromium (Cr), titanium (Ti), or alloys thereof, but the present disclosure is not limited thereto.

(55) Each of the external electrodes **410** and **420** may be formed to have a multilayer structure. For example, each of the external electrodes **410** and **420** may include a first metal layer disposed on the body **100**, a second metal layer disposed on the first metal layer, and a third metal layer disposed on the second metal layer. The first metal layer may be formed by vapor deposition such as sputtering, electroplating, or a paste applying method. The paste applying method means that a conductive paste, including a conductive particle such as copper (Cu) and/or silver (Ag) and a thermosetting resin, is applied to the surface of the body **100** and is then cured. Each of the second and third metal layers may be formed by electroplating. As a non-limiting example, the first metal layer may be a copper plating layer, the second metal layer may be a nickel plating layer, and the third metal layer may be a tin plating layer, but the present disclosure is not limited thereto.

(56) Accordingly, the coil component **1000** according to the present embodiment may prevent plating spreading on the upper and lower surfaces of the body **100** to prevent short-circuits between the external electrodes **410** and **420** when the external electrodes **410** and **420** are plated. In addition, the first cover layer **121** may be additionally provided on the upper surface of the active

portion **110** while significantly reducing a distance between the upper surface of the active portion **110** and the coil portion **300** to reduce an overall thickness of the component, so that a margin of the component in the thickness direction T may be secured.

(57) FIG. **4** is a schematic perspective view of another coil component according to an exemplary embodiment. FIG. **5** is a cross-sectional view taken along line of FIG. **4**, and FIG. **6** is a cross-sectional view taken along line IV-IV' of FIG. **4**. For ease of description, in FIG. **5**, an internal structure of the coil component is illustrated based on the cross-section taken along line III-III' of FIG. **4**.

(58) Referring to FIGS. **1** to **3** and FIGS. **4** to **6**, a difference between a coil component **2000** according to the present embodiment and the coil component **1000** according to an exemplary embodiment exists in first and second cover portions **121** and **122** and external electrodes **410** and **420**. Therefore, the present embodiment will be described while focusing on only the first and second cover portions **121** and **122** and the external electrodes **410** and **420**. The descriptions of the exemplary embodiment may be equivalently applied to the other components of the present embodiment.

(59) Referring to FIGS. **4** to **6**, a first cover portion **121** and a second cover portion **122**, applied to the coil component **2000** according to the present embodiment, may be disposed on only a portion of each of an upper surface and a lower surface of an active portion in a length direction L. Specifically, the first cover portion **121** extends continuously between edges of an upper surface of the active portion **110** in a width direction W, but does not cover both end portions of the upper surface of the active portion **110** in the length direction because the first cover portion **121** is disposed in only a central portion of the upper surface of the active portion **110** in the length direction. The second cover portion **122** extends continuously between edges of a lower surface of the active portion **110** in the width direction, but does not cover both end portions of the lower surface of the active portion **110** in the length direction because the second cover portion **122** is disposed in only a central portion of the lower surface of the active portion **110** in the length direction. As a result, each of the pad portions **411** and **421** and the band portions **412** and **422** of the external electrodes **410** and **420** is in contact with the lower and upper surfaces of the active portion **110**. For example, the first cover portion **121** may cover only a space, in which the pad portions **411** and **421** are spaced apart from each other, of the lower surface of the active portion **110**, and the second cover portion **122** may cover a space, in which the pad portions **411** and **421** are spaced apart from each other, of the lower surface of the active portion **110**. In the present embodiment, since the first and second cover portions **121** and **122** are disposed in only some regions of the upper and lower surfaces of the active portion **110**, an entire thickness of the component may be reduced. In addition, since the first and second cover portions **121** and **122** are disposed in the space in which the band portions **413** and **423** are spaced apart from each other and the space in which the pad portions **411** and **421** are spaced apart from each other, short-circuits between the electrodes **410** and **420** may be prevented when the external electrodes **410** and **420** are formed through a plating process.

(60) FIG. **7** is a schematic perspective view of another coil component according to an exemplary embodiment. FIG. **8** is a cross-sectional view taken along line V-V' of FIG. **7**, and FIG. **9** is a cross-sectional view taken along line VI-VI' of FIG. **7**. For ease of description, in FIG. **8**, an internal structure of the coil component is illustrated based on the cross-section taken along line V-V' of FIG. **7**.

(61) Referring to FIGS. **1** to **3** and FIGS. **7** to **9**, a difference between a coil component **3000** according to the present embodiment and the coil component **1000** according to an exemplary embodiment exists in lead-out patterns **331** and **332**. Therefore, the present embodiment will be described while focusing on only the lead-out patterns **331** and **332**. The descriptions of the exemplary embodiment may be equivalent.

(62) Referring to FIGS. **7** to **9**, in a coil portion **300** applied to the coil component **3000** according

to the present embodiment, the lead-out patterns **331** and **332** may be exposed to a lower surface of the active portion **110**. Specifically, the first lead-out pattern **331** and the second lead-out pattern **332** may be exposed to the lower surface of the active portion **110** in such a manner that they are spaced apart from each other, but may not be exposed to the other surfaces of the body **100**. Thus, the coil component **3000** according to the present embodiment may easily implement a lower electrode structure of a component.

(63) In the present embodiment, external electrodes **410** and **420** are formed to have, for example, an overall L shape, as illustrated in FIGS. **7** to **9**. Specifically, the first external electrode **410** may include a first pad portion **411**, disposed on the lower surface of the active portion **110**, and a first extension portion **412** extending from a first pad portion **411** to the first surface of the body **100**. The second external electrode **420** may include a first pad portion **411**, disposed on the lower surface of the active portion **110**, and a first extension portion extending from the first pad portion **411** to the first surface of the body **100**. However, since the shapes of the external electrodes **410** and **420** illustrated in FIGS. **7** to **9** are only exemplary, the present embodiment is not limited thereto.

(64) As described above, a coil component, in which a coil portion is disposed to be perpendicular to a lower surface of a body, may readily prevent short-circuits between external electrodes.

(65) In addition, a coil component, in which a coil portion is disposed to be perpendicular to a lower surface of a body, may increase an effective volume of a magnetic material.

(66) While exemplary embodiments have been shown and described above, it will be apparent to those skilled in the art that modifications and variations could be made without departing from the scope of the present disclosure as defined by the appended claims.

Claims

1. A coil component comprising: a body having an upper surface and a lower surface opposing each other in a thickness direction of the body; a support substrate disposed in the body and having one surface perpendicular to the lower surface of the body; a coil portion disposed on at least one surface of the support substrate; and a first external electrode and a second external electrode disposed on the lower surface of the body to be spaced apart from each other and respectively connected to the coil portion, wherein the body includes an active portion, covering the coil portion, and first and second cover portions, respectively disposed on an upper surface and a lower surface of the active portion opposing each other in the thickness direction, the active portion includes a first magnetic metal particle, a second magnetic metal particle having a smaller diameter than the first magnetic metal particle, and a first insulating resin, and each of the first and second cover portions includes a third magnetic metal particle, having a smaller diameter than the first magnetic metal particle, and a second insulating resin.
2. The coil component of claim 1, wherein at least a portion of the first magnetic metal particle is cut to be exposed to one of the upper and lower surfaces of the active portion.
3. The coil component of claim 1, wherein the second magnetic metal particle and the third magnetic metal particle have the same diameter.
4. The coil component of claim 1, wherein the body further includes a third cover portion and a fourth cover portion, respectively disposed on both side surfaces of the active portion opposing each other in a width direction of the body, and each of the third cover portion and the fourth cover portion includes a fourth magnetic metal particle, having a smaller diameter than the first magnetic metal particle, and a third insulating resin.
5. The coil component of claim 4, wherein the fourth magnetic metal particle and the third magnetic metal particle have the same diameter.
6. The coil component of claim 4, wherein the first and second cover portions respectively extend from the upper surface and the lower surface of the active portion to cover at least a portion of each

of an upper surface and a lower surface of each of the third and fourth cover portions.

7. The coil component of claim 6, wherein the upper surface of the active portion and the upper portion of each of the third and fourth cover portions are coplanar with each other.

8. The coil component of claim 6, wherein the body further has one end surface and the other end surface opposing each other in a length direction of the body, the first and second external electrodes respectively includes pad portions, disposed on the lower surface of the active portion to be spaced apart from each other, and extension portions extending to the one end surface and the other end surface of the body, each of the pad portions is in contact with the lower surface of the active portion, the second cover portion is disposed in a separation space between the pad portions of the lower surface of the active portion, the first cover portion covers an entire upper surface of the active portion, and each of the extension portions is in contact with a side surface of the first cover portion and does not extend upwardly of the upper surface of the first cover portion.

9. The coil component of claim 1, wherein the body further has one end surface and the other end surface opposing each other in a length direction of the body, and the first and second external electrodes respectively include pad portions disposed on the lower surface of the active portion to be spaced apart from each other, extension portions extending from the pad portions to the one end surface and the other end surface of the body, and band portions disposed on the upper surface of the active portion to be spaced apart from each other.

10. The coil component of claim 9, wherein the first and second cover portions cover entire upper and lower surfaces of the active portion, respectively, and the band portions and the pad portions are disposed on the first and second cover portions, respectively.

11. The coil component of claim 9, wherein each of the band portions and the pad portions is in contact with the upper surface or the lower surface of the active portion, and the first cover portion is disposed in a separation space between the band portions of the upper portion of the active portion, and the second cover portion is disposed in a separation space between the pad portions of the lower surface of the active portion.

12. A coil component comprising: a body having a lower surface; a support substrate disposed in the body and having one surface perpendicular to the lower surface of the body; and a coil portion disposed on at least one surface of the support substrate, wherein the body includes an active portion, covering the coil portion, and first to fourth cover portions, respectively disposed on an upper surface of the body, the lower surface of the body, one side surface of the body, and the other side surface of the body, each of the active portion and the first to fourth cover portions includes magnetic metal particles and an insulating resin, and a filling rate of the magnetic metal particles of the active portion is greater than a filling rate of the magnetic metal particles of each of the first to fourth cover portions.

13. The coil portion of claim 12, wherein the first cover portion has a boundary with the upper surface of the active portion, and the second cover portion has a boundary with the lower surface of the active portion, and at least a portion of the magnetic metal particle is cut to be exposed to the boundary.

14. A coil component comprising: a body having an upper surface and a lower surface opposing each other; a coil portion disposed in the body vertically with respect to the lower surface of the body; and a first external electrode and a second external electrode respectively including pad portions disposed on the lower surface of the body to be spaced apart from each other and connected to the coil portion, wherein an average size of metal particles dispersed in a portion, in contact with one of the pad portions, of the body, is less than an average size of metal particles dispersed in another portion, spaced apart from outer surfaces of the body, of the body.

15. The coil component of claim 14, wherein a metal particle in the another portion of the body include a flat surface and a curved surface intersected by the flat surface.

16. The coil component of claim 14, wherein the body further has one end surface and the other end surface opposing each other, and the first and second external electrodes respectively further

include extension portions extending from the pad portions to the one end surface and the other end surface of the body, and band portions disposed on the upper surface of the active portion to be spaced apart from each other.

17. The coil component of claim 16, wherein an average size of metal particles dispersed in a portion, in contact with one of the band portions, of the body, is less than the average size of metal particles dispersed in the another portion, spaced apart from the outer surfaces of the body, of the body.

18. The coil component of claim 14, wherein the body further has one end surface and the other end surface opposing each other, and the first and second external electrodes respectively further includes extension portions extending to the one end surface and the other end surface of the body.

19. The coil component of claim 18, wherein an average size of metal particles dispersed in a portion, in contact with one of the extension portions, of the body, is less than the average size of metal particles dispersed in the another portion, spaced apart from the outer surfaces of the body, of the body.

20. The coil component of claim 14, wherein a metal particle in the body including a flat surface and a curved surface intersected by the flat surface is spaced apart from the outer surfaces of the body.

21. The coil component of claim 14, wherein a winding axis of the coil portion is parallel to the lower surface.
